THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 6–10: **We Distribute, Yet Things Multiply, Proportional Reasoning-1, Fractions in Disguise, The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 53	$x^2+y^2=(x+y)^2-2xy$
2	**(b)** 8 : 12 : 21	Make the middle terms equal using their LCM
3	**(c)** 1%	Loss % = (p²/100) = 10²/100
4	**(d)** 120	Legs 20 and 48 since $5^2+12^2=13^2$
5	**(b)** 24	Let ages be $3x$, $5x$: $(3x+8):(5x+8)=2:3$ gives $x=8$
6	**(c)** 49	$x^2+y^2=(x-y)^2+2xy$
7	**(b)** 24 years	Let the ages be $3x$ and $2x$; solving $(3x+4)\times5=(2x+4)\times7$ gives $x=8$
8	**(a)** 60500	50000×(1+10/100)²
9	**(b)** 972	Sides 27 and 36; area $=27\times36$
10	**(d)** 4	Rate $=\frac{1}{12}+\frac{1}{6}$, time $=\frac{12\times6}{18}$
11	**(d)** 24000	$(512+488)(512-488)$
12	**(b)** ₹ 600	Convert the reciprocals to whole-number ratio using the LCM
13	**(c)** 40500	50000×(1−10/100)²
14	**(c)** 15, 112, 113	Use $(n,\frac{n^2-1}{2},\frac{n^2+1}{2})$ for odd n and $(n,\frac{n^2}{4}-1,\frac{n^2}{4}+1)$ for even n
15	**(d)** 104	$13\times360/45$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$x^2+y^2=(x+y)^2-2xy$ (1) $=144-70=74$ (1).	1+1
17	Perimeters $3:7$ (1); areas $9:49$ (1).	1+1
18	Final = (1+25/100)(1−10/100) (1) = 112.5% of original, so more by 12.5% (1).	1+1
19	Let legs be $3x$, $4x$; hypotenuse $=5x=35$ (1); $x=7$, legs 21 cm and 28 cm (1).	1+1
20	Take $x=3k$, $y=4k$ (1); ratio $=23k:8k=23:8$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$x^2+y^2=(x-y)^2+2xy=16+42=58$ (1.5); $(x+y)^2=(x-y)^2+4xy=16+84=100$ (1.5).	1½+1½
22	Let ages be $3x$ and $1x$ (1); $(3x+4)\times3=(1x+4)\times7$ gives $x=8$ (1); ages 24 and 8 years (1).	1+1+1
23	CP1 = 1980×100/110 = 1800 (1); CP2 = 1980×100/90 = 2200 (1); total CP − total SP = ₹ 40, a loss (1).	1+1+1
24	Foot to first wall $=\sqrt{37^2-35^2}=12$ m (1); foot to second wall $=\sqrt{37^2-12^2}=35$ m (1); width $=47$ m (1).	1+1+1
	OR: Remaining provisions $=(45-15)\times240=7200$ man-days (1); men left $=180$ (1); days $=40$ (1).	1+1+1
25	$(3x+2)^2=9x^2+12x+4$ (1); $(3x-2)^2=9x^2-12x+4$ (1); difference $=24x$ (1).	1+1+1
	OR: Total work $=360$ man-days; done $=120$ (1); left $=240$ (1); days $=240\div20=12$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Area $=(3x+2)(3x-2)=9x^2-4$ (2); at $x=5$: 221 m² (1); square $=9x^2$, difference $=4$ m² (2).	2+1+2
	OR: Investment × time: 12, 12, 15 (1); ratio $4:4:5$ (1); one part ₹ 100 (1); shares ₹ 400, ₹ 400, ₹ 500 (2).	1+1+1+2
27	Year 1: A1 = ₹ 12000, interest ₹ 2000 (1); year 2: A2 = ₹ 14400, interest ₹ 2400 (2); year 3: A3 = ₹ 17280, interest ₹ 2880 (1); total ₹ 7280 = A3 − P (1).	1+2+1+1
	OR: $AC^2=9^2+12^2=225$, $AC=15$ cm (1); $AC^2+CD^2=225+400=625=DA^2$ so $\angle ACD=90^\circ$ (2); area $=\frac{1}{2}\times9\times12+\frac{1}{2}\times15\times20=204$ cm² (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapters 6–10: **We Distribute, Yet Things Multiply, Proportional Reasoning-1, Fractions in Disguise, The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 7	Square both sides
2	**(b)** 9 days	Find the work left in man-days and divide by the new number of men
3	**(d)** −4%	(1+20/100)(1−20/100) − 1
4	**(d)** 5280	Side 73; half other diagonal 55; area $= \frac{1}{2}d_1d_2 = 5280$
5	**(a)** 140	Combine to $a : b : c$ first
6	**(d)** 25	$p+q=a$, $pq=b$
7	**(c)** 5 : 4	Time is inversely proportional to speed
8	**(a)** 2500	A = P(1+20/100)², CI = A − P
9	**(d)** 17	Foot distances $\sqrt{13^2-12^2}=5$ and $\sqrt{13^2-5^2}=12$ add up to the width
10	**(c)** 24	Remaining food $= (30-10)\times 120$ man-days
11	**(d)** 141	25^2-22^2
12	**(c)** 32	Adding the denominators keeps the ratio
13	**(b)** 26%	(1+40/100)(1−10/100) − 1
14	**(a)** 92	Rectangle property $PA^2+PC^2=PB^2+PD^2$
15	**(c)** 6	New water $= 36\times 1/2 = 18$ L

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$(612+388)(612-388) = 1000\times 224$ (1) $= 224000$ (1).	1+1
17	Take $x=7k$, $y=3k$ (1); $(10k):(4k)=5:2$ (1).	1+1
18	Marked price = ₹ 2100 (1); SP = ₹ 1890 (1).	1+1
19	Triplet $(19, 180, 181)$ (1); check $19^2+180^2=32761=181^2$ (1).	1+1
20	Part filled in 1 hour $= \frac{1}{18}+\frac{1}{9}=\frac{27}{162}$ (1); time $= 6$ hours (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	V = P(1−r/100)³ (1) = 500000×90/100 three times (1) = ₹ 364500 (1).	1+1+1
22	Side $= 73$ cm; half-diagonal $= 48$ cm (1); other half $= \sqrt{73^2-48^2}=55$ cm, other diagonal 110 cm (1); area $= \frac{1}{2}\times 96\times 110 = 5280$ cm² (1).	1+1+1
23	Let milk $= 3k$, water $= 2k$ (1); $(2k+10)/(3k)=4/3$ gives $k=5$ (1); milk $= 15$ L, water $= 10$ L (1).	1+1+1
24	$a^2+b^2=(10)^2-2(24)=52$ (1.5); $(a-b)^2=a^2+b^2-2ab=52-48=4$ (1.5).	1½+1½
	OR: LCM of 2, 3, 4 is 12; ratio $6:4:3$ (2); parts ₹ 1200, ₹ 800, ₹ 600 (1).	2+1
25	MP = ₹ 1200 (1); SP = ₹ 960 (1); gain/loss % = 20% (1).	1+1+1
	OR: East distance $= 24$ km, south distance $= 45$ km (1); distance$^2 = 576+2025=2601$ (1); distance $= 51$ km (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Let ages be $5x$, $8x$ (1); $(5x+6):(8x+6)=2:3$ gives $x=6$ (2); ages 30 and 48 years (1); 6 years ago: 24 and 42, ratio $4:7$ (1).	1+2+1+1
	OR: $xy=\dfrac{(13)^2-97}{2}=36$ (2); $(x-y)^2=97-2(36)=25$ (2); $x-y=5$ (1).	2+2+1
27	Milk 40 L, water 20 L (1); in 15 L removed: milk 10 L, water 5 L (1); milk left 30 L (1); water now $20-5+15=30$ L (1); ratio $1:1$ (1).	1+1+1+1+1

	OR: Let CP = x. SP = x(1+50/100)(1−20/100) = 1.2x (2); profit = 0.2x = 240 (2); x = ₹ 1200 (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 3

Chapters 6–10: **We Distribute, Yet Things Multiply, Proportional Reasoning-1, Fractions in Disguise, The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 47	$n^2-(n-1)^2=2n-1$
2	**(c)** 4 : 9	Area $=$ side^2
3	**(c)** 7500	Difference = P(R/100)²
4	**(b)** 7	$\sqrt{2^2+3^2+6^2}=7$
5	**(b)** 18900	Wage per worker per day $=$ ₹150
6	**(b)** $ab=0$	Compare with $a^2+2ab+b^2$
7	**(d)** 27	$x=\frac{9\times12}{4}$
8	**(b)** 50%	SI = 2P so 2P = P×R×T/100 and R = 200/T
9	**(b)** 120	Legs 24 and 45 since $8^2+15^2=17^2$
10	**(b)** 12	Let ages be $2x$, $5x$: $(2x+9):(5x+9)=7:13$ gives $x=6$
11	**(a)** 65	$x^2+y^2=(x+y)^2-2xy$
12	**(c)** 9 : 12 : 10	Make the middle terms equal using their LCM
13	**(b)** 4%	Loss % = (p²/100) = 20²/100
14	**(c)** 540	Sides 15 and 36; area $=15\times36$
15	**(c)** 6	Rate $=\frac{1}{15}+\frac{1}{10}$, time $=\frac{15\times10}{25}$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$p+q=8$, $pq=15$ (1); $p^2+q^2=8^2-2(15)=34$ (1).	1+1
17	Ratio $\frac{1}{2}:\frac{1}{3}=3:2$ (1); parts ₹ 720 and ₹ 480 (1).	1+1
18	A = 2500(120/100)² = ₹ 3600 (1); A/P = 1.44 (1).	1+1
19	$PA^2+PC^2=PB^2+PD^2$ (1); $PD^2=64+100-36=128$ (1).	1+1
20	Let ages be $7x$, $2x$ (1); solve $(7x+6):(2x+6)=8:3$ to get $x=6$, ages 42 and 12 years (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Students per degree $=12/40$ (1); total $=108$ (1); Maths $=24$ students (1).	1+1+1
22	$x^2+y^2=(x-y)^2+2xy=36+80=116$ (1.5); $(x+y)^2=(x-y)^2+4xy=36+160=196$ (1.5).	1½+1½
23	Let ages be $5x$ and $2x$ (1); $(5x+6)\times1=(2x+6)\times2$ gives $x=6$ (1); ages 30 and 12 years (1).	1+1+1
24	CP1 = 2640×100/120 = 2200 (1); CP2 = 2640×100/80 = 3300 (1); total CP − total SP = ₹ 220, a loss (1).	1+1+1
	OR: Foot to first wall $=\sqrt{53^2-45^2}=28$ m (1); foot to second wall $=\sqrt{53^2-28^2}=45$ m (1); width $=73$ m (1).	1+1+1
25	Remaining provisions $=(40-10)\times180=5400$ man-days (1); men left $=135$ (1); days $=40$ (1).	1+1+1
	OR: $(4x+5)^2=16x^2+40x+25$ (1); $(4x-5)^2=16x^2-40x+25$ (1); difference $=80x$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Floor diagonal$^2=1^2+2^2=5$ (1), floor diagonal $=\sqrt{5}$ m (1); rod$^2=5+2^2=9$ (2); rod $=3$ m (1).	1+1+2+1
	OR: Work left after 10 days $=(30-10)\times24=480$ man-days (2); workers now $=16$ (1); more days $=30$ (1); total $=40$ days, delay $=10$ days (1).	2+1+1+1
27	Area $=(4x+3)(4x-3)=16x^2-9$ (2); at $x=6$: 567 m² (1); square $=16x^2$, difference $=9$ m² (2).	2+1+2
	OR: Investment × time: 12, 12, 8 (1); ratio 3 : 3 : 2 (1); one part ₹ 150 (1); shares ₹ 450, ₹ 450, ₹ 300 (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND